

# Marketing Analytics

Chapter 1 · Marketing Analytics as Decision-Making

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# Today's session

## Part A — Why we are here

- 1.1 Why marketing runs on data
- 1.2 A \$50 million question

## Part B — The framework of the course

- 1.3 Four kinds of questions
- 1.4 Correlation, causation & what data can show
- 1.5 The cast of variables
- 1.6 Why randomization works
- 1.7 Between the poles: quasi-experiments

## Part C — Making it work

- 1.8 From business problem to research design
- 1.9 The analytics workflow (and AI assistants)
- 1.10 The course at a glance

**One promise for today:** no formulas, no R output, no tests. Concepts and pictures only — the machinery follows in later sessions.

# Who I am...



- **Professor** of Marketing, Vienna University of Economics and Business, Institute for Retailing & Data Science, since 2021
- **Assistant Professor** of Marketing, Vienna University of Economics and Business, Institute for Interactive Marketing & Social Media, 2015–2020
- **Ph.D.** in Marketing, University of Hamburg, 2010–2014
- **Dissertation topic:** “Information Technology and Online Content Distribution”
- **Universal Music Group:** Business Analyst & Digital Marketing Manager, 2008–2010
- **Research interests:** multi-channel distribution, online content distribution, digital business models, marketing modeling, e-commerce policies, information goods, targeting & consumer privacy ...

## Contact

Univ.Prof. Dr. Nils Wlömert

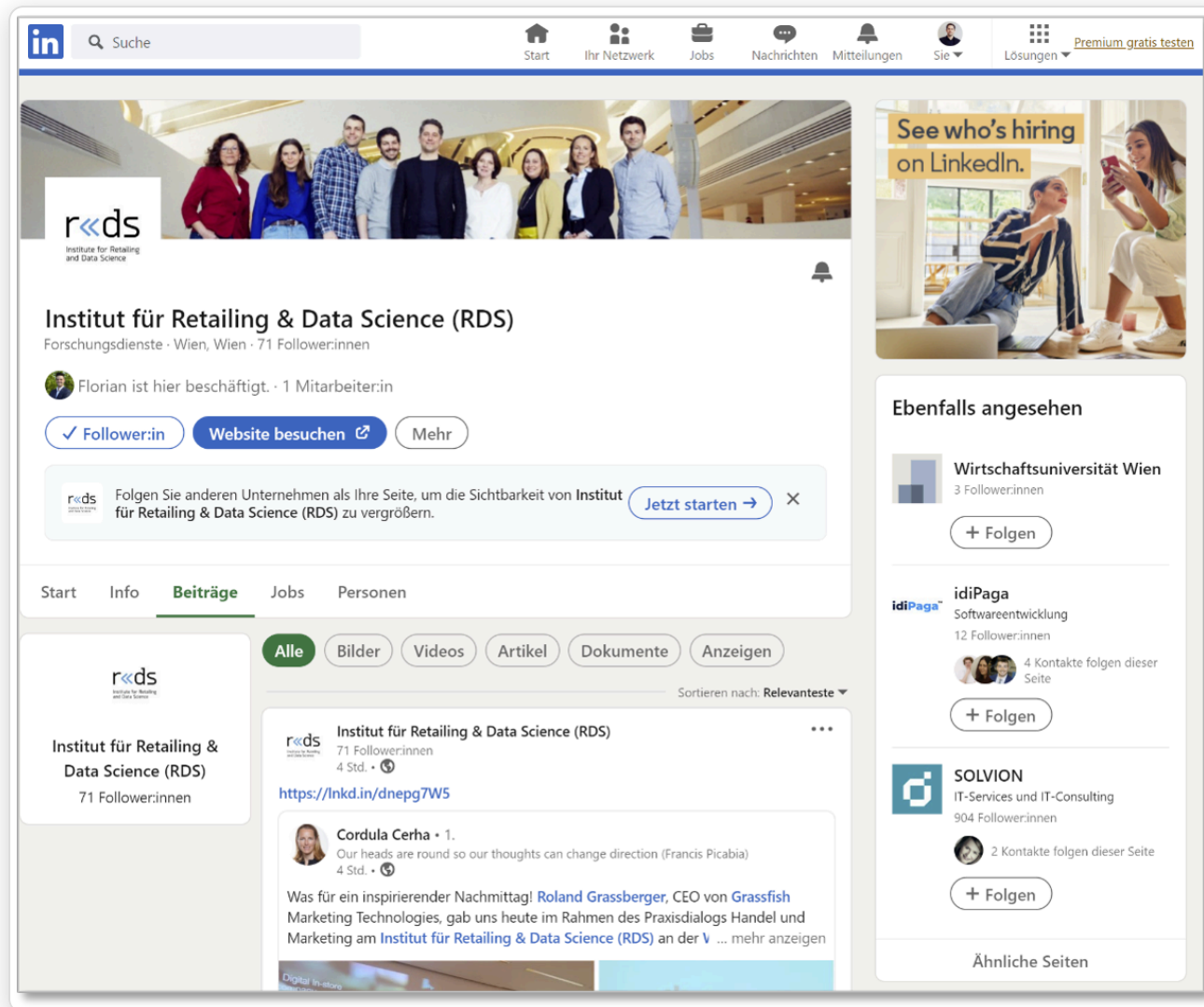
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# 1.1 Why marketing runs on data

# The manager's job description has changed

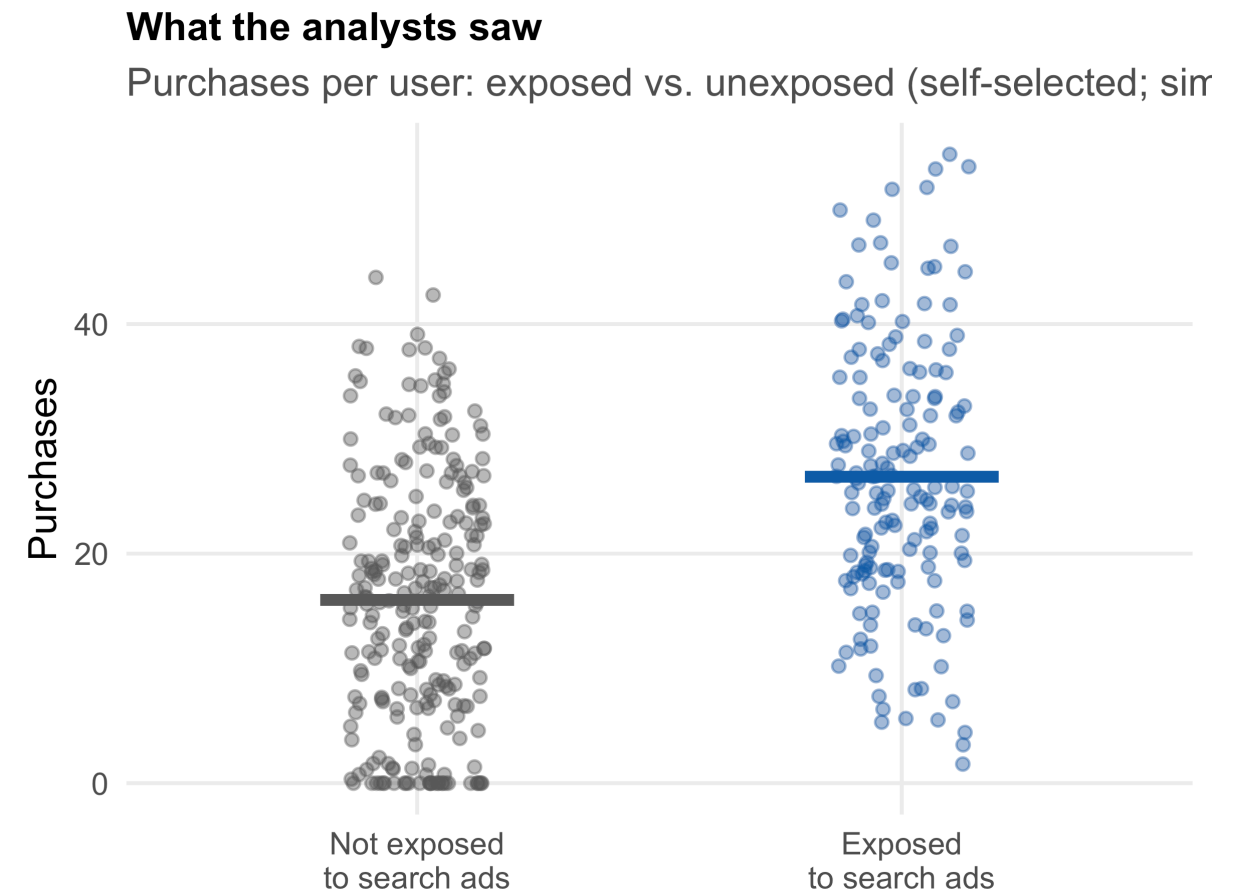
- Every purchase, click, swipe, and like leaves a trace — firms sit on more behavioral data than they can use
- Surveys of CMOs consistently rank **data & analytics competence** among the top changes to the marketing role
- The bottleneck is no longer *collecting* data — it is **drawing valid conclusions** from it

This course is **not** about producing numbers. It is about knowing **which analysis answers which managerial question** — and executing it credibly in R.

# 1.2 A \$50 million question

# eBay, 2012: does paid search advertising pay off?

- eBay spent ~ \$50M **per year** on paid search ads (Google, Bing, Yahoo)
- The analytics team examined the data: users **exposed to ads bought far more** than users who weren't
- Measured “returns” on brand-keyword ads looked spectacular



# How the “returns” were measured: click attribution

1. A user types “**ebay guitar**” into Google. The top result is eBay’s **paid** link; directly beneath it sits eBay’s **organic** link.
2. She clicks the paid link. The URL carries a **tracking parameter**, so eBay’s systems log: this visit arrived via a paid search click, on this keyword, at this cost-per-click.
3. Within the **attribution window** (say, some days), she buys a €150 guitar. The attribution system **credits the €150 to paid search** — the sale is “attributed” because it followed an ad click.
4. Aggregate over millions of users: **attributed revenue ÷ ad spend** = a spectacular return, per keyword, in a dashboard, updated daily.

**Discuss (5 min):** Would you base the decision to invest in keyword advertising on this analysis? What could be potential problems with it?

# What the experiment revealed

- **Brand keywords** (“ebay shoes”): ads halted entirely on Yahoo! & Microsoft, Google kept as comparison → **~99.5% of the paid clicks simply returned via the free organic link below the ad**
- **Non-brand keywords** (“used guitar”): all ads suspended in a **randomly selected ~30% of US regions for 60 days**, change in sales compared against the other regions
- Effect only among **new & infrequent users**; frequent shoppers — most of the clicks and cost — were unmoved
- The naive analysis had credited the shoppers’ **intent** to the ads; true average returns: **negative**

Smart people, huge data, careful computation — and a conclusion that was off by tens of millions of dollars. **The problem was not the math. It was the comparison.**

# The comparison that matters

**The causal question:** *What would eBay's sales have been **without** the ads — everything else unchanged?*

|                        | Observed                 | Needed                            |
|------------------------|--------------------------|-----------------------------------|
| Group A                | Exposed users' sales     | Exposed users' sales              |
| Group B                | Unexposed users' sales   | <b>The same users</b> , unexposed |
| Difference measures... | Exposure + <b>intent</b> | Effect of exposure only           |

The “needed” Group B is a **counterfactual** — a world we can never observe directly. All of causal inference is about **constructing a credible stand-in** for it. This single idea organizes the entire course.

# 1.3 Four kinds of questions

# Every analytics task answers one of four questions

| # | Question                | Type         | Marketing examples                                                 |
|---|-------------------------|--------------|--------------------------------------------------------------------|
| 1 | What is going on?       | Descriptive  | Segment sizes, churn rate, funnel drop-off, satisfaction tracking  |
| 2 | What will happen?       | Predictive   | Which customers will churn next quarter? Demand next month?        |
| 3 | What happens if we act? | Causal       | Does the discount <i>cause</i> extra sales? Did the campaign work? |
| 4 | What should we do?      | Prescriptive | Set the price; allocate budget across channels                     |

**Prescription requires causation.** To choose an action you must know what each action would *cause* — i.e., compare counterfactuals. Prediction alone is not enough... and confusing 2 with 3 is precisely the eBay mistake.

# Prediction $\neq$ causation: same data, different questions

“Customers who see our newsletter churn 40% less.”

As a predictive statement ✓

Newsletter readership *predicts* retention — useful for **targeting**: flag non-readers as churn risks.

*Valid even if the newsletter does nothing.*

As a causal statement ✗ (so far)

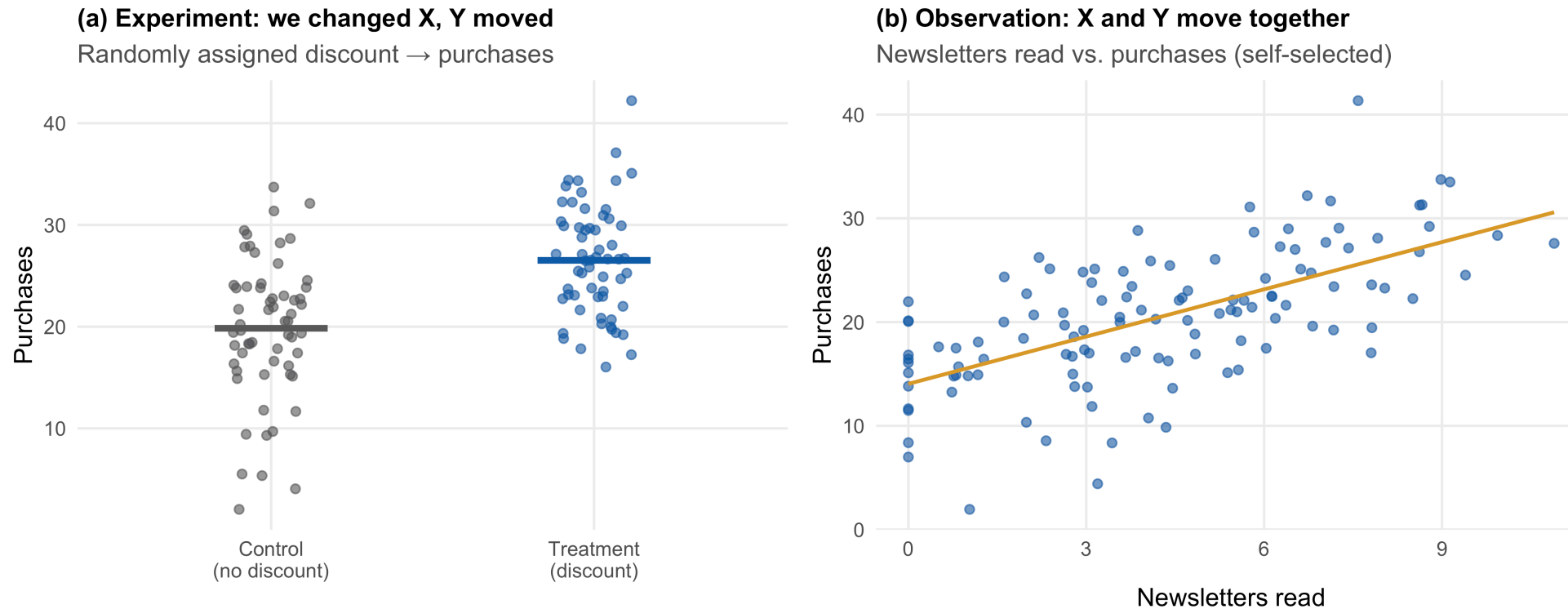
“Send more newsletters  $\rightarrow$  churn drops”?  
Only if readership *causes* retention — but engaged customers **both** read newsletters **and** stay.

*Requires a counterfactual comparison.*

**Classify these (2 min, neighbor discussion):** (a) “Which ad creative will get the highest CTR?” (b) “How large is our Gen-Z segment?” (c) “Should we raise the price by 5%?”

# 1.4 Correlation, causation & what data can show

# Two pictures of “X relates to Y”



The two pictures can look **equally convincing** — yet only (a) licenses a causal conclusion. The difference is invisible in the plot: it lies in **how the data came to be** (the *data-generating process*).

# Three ways to get the same scatter plot

# Reverse causality: who is causing whom?

Observed: regions with higher ad spend show higher sales.



*Story 1:* advertising drives sales.

*Story 2:* budgets are **set as a % of sales** — sales drive ad spend.

Both stories produce identical data. Marketing budget rules make reverse causality a **routine** hazard in marketing data, not an exotic one.

# 1.5 The cast of variables

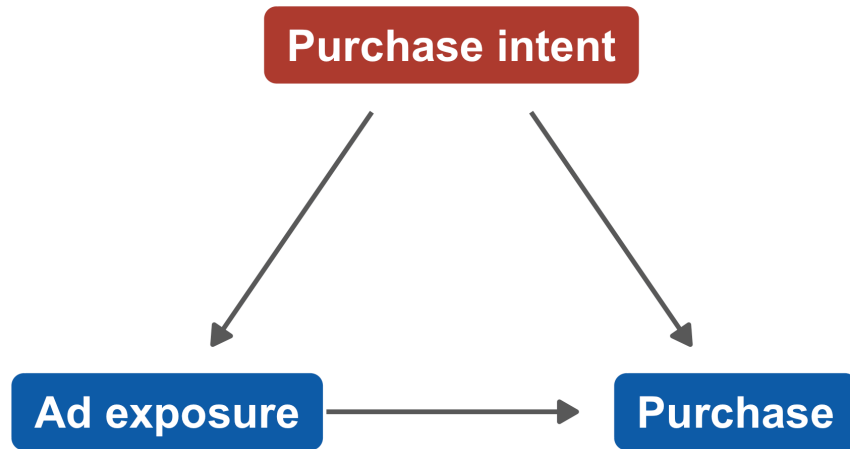
# The two lead roles



- **Outcome / dependent variable (DV):** the quantity the manager cares about
- **Treatment / independent variable (IV):** the lever the manager can pull

Defining these two precisely — *whose* purchases, over *what* horizon, *which* discount — is half of research design (§1.8).

# The villain: the confounder



- Causes **both** treatment and outcome
- Creates correlation **without** causation — panel 2 of our gallery
- **The eBay confounder:** purchase intent → people search for “eBay” → see the ad → and buy anyway
- Whole toolkits (Ch. 5–6) exist to deal with confounders in observational data

# Supporting roles: mediator and moderator



## Mediator — the *how*

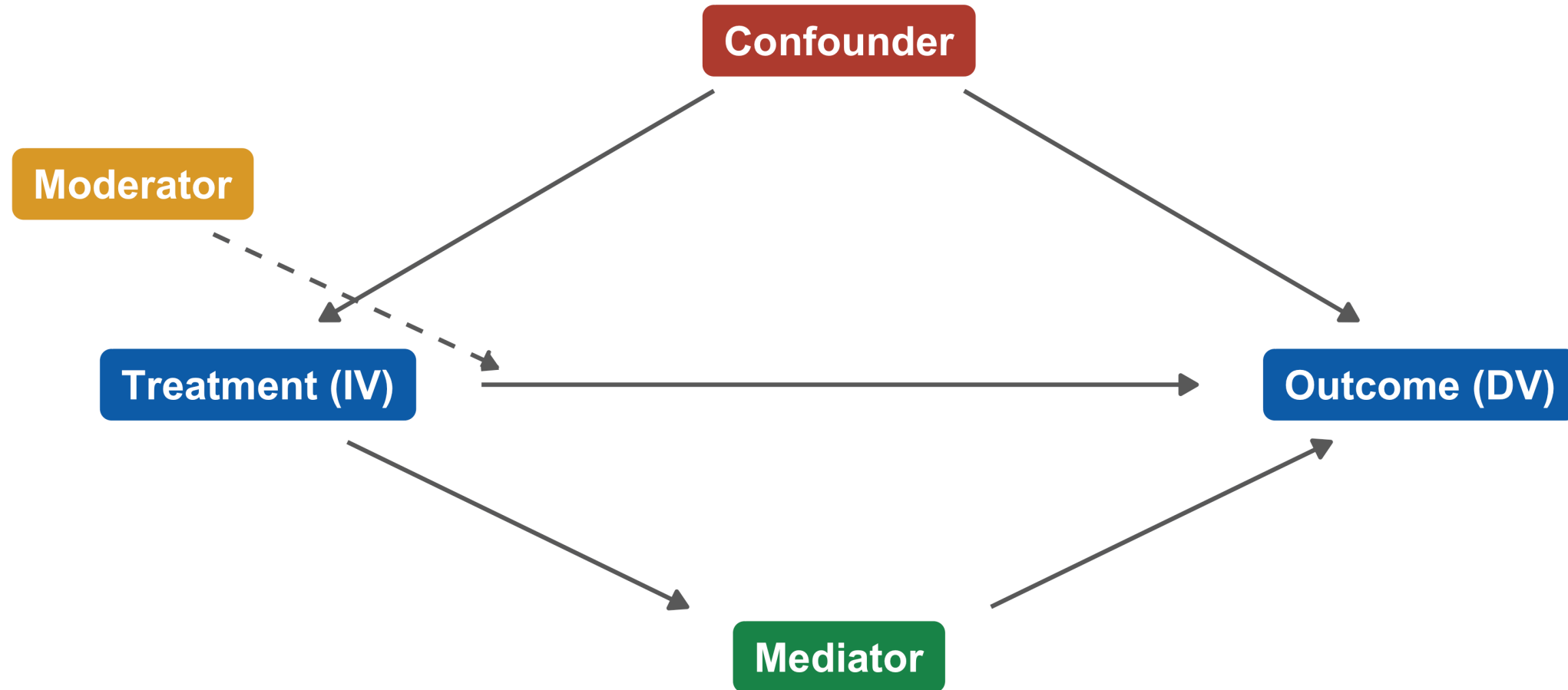
Lies **on the path**: the mechanism through which X works.  
Why care? It tells you **what to optimize** (creative that builds awareness).

## Moderator — the *for whom / when*

Points **at the arrow**: changes the **strength** of the effect.  
“The discount works for new customers, not loyal ones.”

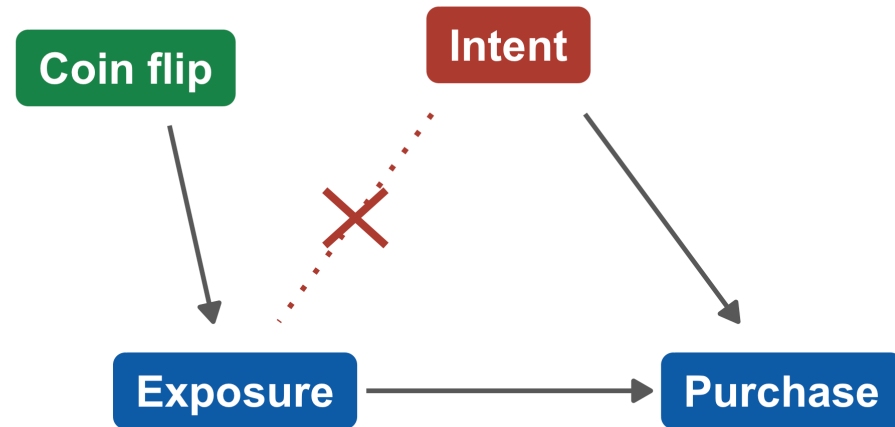
Today: **recognize** the roles. Chapter 5 teaches how to **handle** each one — and shows that mishandling them (e.g., “controlling” for a mediator) can wreck an analysis.

# The full cast around one causal question



# 1.6 Why randomization works

# Randomization cuts every arrow into the treatment

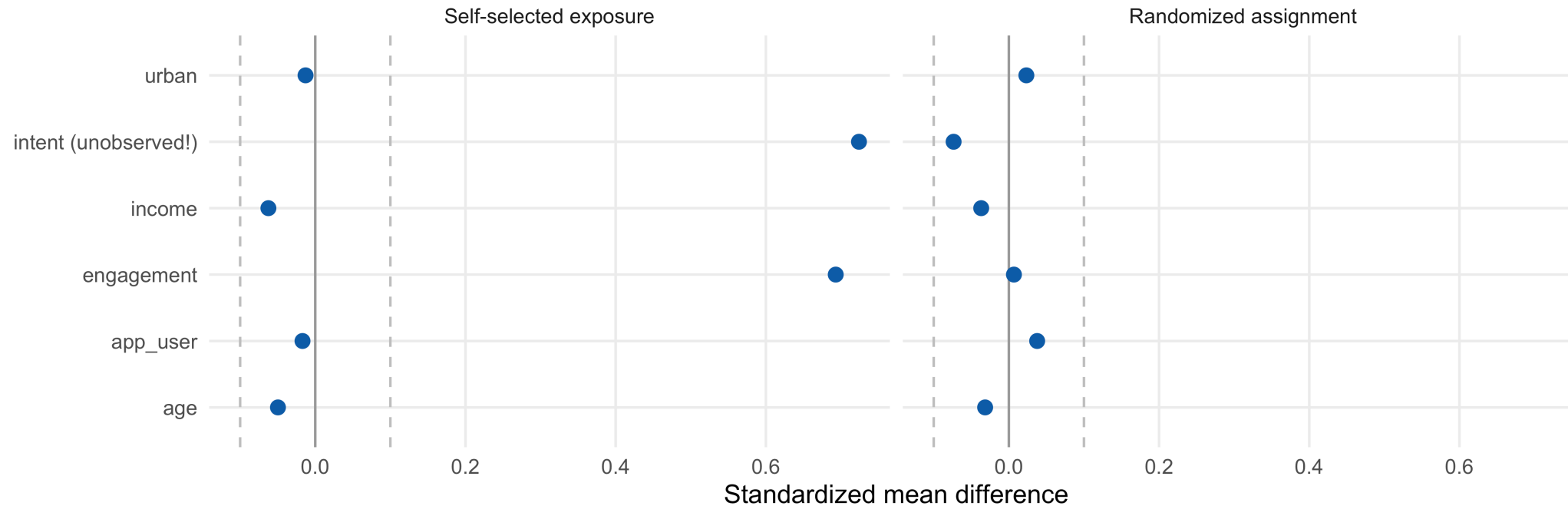


- Assignment now depends **only on the coin** — not on intent, engagement, income, mood, ...
- Confounders (known **and unknown**) are balanced across groups *in expectation*
- The control group becomes a valid **counterfactual**: same kinds of people, minus the treatment
- Differences in outcomes can then be attributed to the treatment (up to chance — quantified in Ch. 3)

# “Balanced — even on what we never measured”

**Covariate balance: exposed vs. unexposed group (simulated, n = 2,000)**

Standardized mean difference per variable · dashed lines: conventional  $\pm 0.1$  band



Left: the groups differ systematically — including on **intent**, which we (realistically) never observe. Right: one coin flip balances everything at once, **without knowing what needs balancing**. That is the magic randomization buys.

# The price of observational data

Causal conclusions from observational data require assuming:

**No unobserved confounders** — every variable that drives both treatment and outcome has been measured and correctly adjusted for.

- This assumption is **untestable in principle**: you cannot check balance on variables you did not observe
- Experiments **buy** the assumption *by design*; observational studies must **argue** for it — with domain knowledge, not statistics
- This asymmetry is the most important sentence of the course: it determines what your data can and cannot claim
- It is *not* a counsel of despair — Ch. 5–6 develop the toolkit for making the argument as credible as possible

# 1.7 Between the poles: quasi-experiments

# Sometimes the world randomizes for you (almost)

## A spectrum of designs:

|                        | Experiment                 | Quasi-experiment                          | Observational                      |
|------------------------|----------------------------|-------------------------------------------|------------------------------------|
| Who assigns treatment? | The researcher (coin flip) | External event / rule                     | The units themselves               |
| Example                | A/B test of a discount     | Platform launches in some countries first | Newsletter readers vs. non-readers |
| Key requirement        | Correct execution          | “As-good-as-random” timing/rule           | <b>No unobserved confounders</b>   |

- Policy changes, staggered rollouts, regional boundaries → treatment variation **not chosen by the units**
- Workhorse method: **difference-in-differences** — compare *changes* in treated vs. untreated markets
- Extremely common in practice (and in marketing research) → dedicated session in **Module 6 (panel data)**

# 1.8 From business problem to research design

# The translation template

**Manager:** *“I want to know whether our loyalty program is worth the money.”*

| Design element     | Your translation                                                                  |
|--------------------|-----------------------------------------------------------------------------------|
| Outcome (DV)       | Customer contribution margin, 12 months                                           |
| Treatment (IV)     | Loyalty program membership                                                        |
| Unit of analysis   | Customer                                                                          |
| Question type      | Causal (→ prescriptive decision)                                                  |
| Likely confounders | Purchase frequency & engagement <i>before</i> joining                             |
| Feasible design?   | Randomize invitations? Staggered regional rollout?<br>Observational + adjustment? |

A precise, falsifiable formulation (“membership increases 12-month margin”) beats a vague aspiration (“the program creates value”) — it dictates data, design, and analysis.

# Your turn (pairs, 10 min)

Translate one of these into the template:

1. “Should we invest more in influencer marketing?”
2. “Is our app driving retention, or do loyal customers just download the app?”
3. “Did last spring’s rebranding damage sales in Germany?”

For each: DV, IV, unit, question type, two plausible confounders, and the **best feasible design** (experiment / quasi-experiment / observational).

# 1.9 The analytics workflow — and AI assistants

# Where AI helps — and where it cannot replace you

| Workflow step               | Nature of the work | AI assistant...                                            |
|-----------------------------|--------------------|------------------------------------------------------------|
| Formulate the question      | <b>Judgment</b>    | can challenge you — cannot know your business              |
| Choose design & method      | <b>Judgment</b>    | suggests options — <i>you</i> own the assumptions          |
| Write & debug R code        | Mechanical         | <b>excellent</b> — use it freely, including in this course |
| Interpret results           | <b>Judgment</b>    | drafts text — but cannot verify the causal claim           |
| Communicate to stakeholders | Mixed              | strong support, needs your framing                         |

**Course policy in one line:** use AI to *accelerate* your learning of the judgment steps, not to *skip* them. Assessments in this course are designed so that the judgment steps are what earns the points.

# 1.10 The course at a glance

# One map for the whole course

| Module                               | Sessions                 | Driving question                                    |
|--------------------------------------|--------------------------|-----------------------------------------------------|
| 1 · Questions, decisions & data      | 05.10.                   | Which question am I really asking? ( <i>today</i> ) |
| 2 · Describing data                  | 12.10.                   | What is going on?                                   |
| 3 · Uncertainty                      | 19.10.                   | How sure can I be?                                  |
| 4 · Learning from experiments        | 19.10. / 27.10.          | Did it work?                                        |
| 5 · Learning from observational data | 02.11. / 09.11. / 16.11. | What if we cannot randomize?                        |
| 6 · Panel data & quasi-experiments   | 30.11.                   | What if the world randomizes for us?                |
| 7 · From estimate to decision        | 07.01.                   | What should we do?                                  |
| Group project coaching               | 23.11. / part of 07.01.  | Your own experiment, end to end                     |
| Final exam (in person)               | 15.12.                   | —                                                   |

**Group project preview:** design a survey **containing a real randomized experiment**, field it, analyze it. Everything from today — counterfactuals, randomization, confounders — becomes something you *build*, not just recognize.

# Learning check — can you spot the reasoning?

1. An analyst reports: “*App users have 2× the lifetime value of non-users — the app clearly pays off.*” Which question type is being **answered**, which is being **claimed**? Name the most plausible confounder.
2. A platform’s advertising dashboard compares purchases of users who clicked an ad vs. users who saw it and didn’t click. Is this an experiment? What role does “interest in the product” play?
3. Your CMO wants to know whether to extend a price promotion nationally after a pilot in Vienna. What design element makes the pilot informative — and what is the counterfactual?
4. For each: newsletter → (*opens* →) purchases; discount effect stronger for students — mediator or moderator?

# Next session

## Module 2 · Describing data — *What is going on?*

- Getting data into R, cleaning it, and not trusting it too early
- Summary statistics and the visualizations that earn their place
- **Before the session:** complete the R onboarding module (installation *or* Posit Cloud) — link on the course site. Any AI assistant is an endorsed study aid for this.

Course book chapter 1 (consolidation reading for today) is online — including all simulated examples from these slides, with their code.